Appendix and Supplementary Materials: ProDA: Projection-based Distribution Alignment for Deep Time Series Forecasting

A Proof/Mathematical Derivation

A.1 Proof for Theorem 1

Proof. We rely on the classical Cramér–Wold theorem [1], which states that a multivariate distribution is uniquely determined by its one-dimensional projections.

We omit the proof and refer to the proof of Theorem I in [1] for details. □

A.2 Proof for Proposition 1

The first result recalls a basic but important property of the Cramér–Wold discrepancy.

Lemma 1 (Metric [3])**.** *D_{CW} is a metric on the space of probability measures.*

This lemma implies that zero Cramér–Wold discrepancy guarantees equality of the underlying distributions (refer to the proof of Theorem 6 in [3]).

Proof. Since the Cramér–Wold discrepancy is a metric on probability measures, $D_{\mathrm{CW}}(\mu, \hat{\mu}) = 0 \iff \mu = \hat{\mu}$.

Firstly, denoted by $\mathbf{x}^{(t,d)}$, $\mathbf{y}^{(t,d)}$, and $\hat{\mathbf{y}}^{(t,d)}$ the realizations of $\mathbf{x}^{(d)}$, $\mathbf{y}^{(d)}$, and $\hat{\mathbf{y}}^{(d)}$ respectively. Define μ and $\hat{\mu}$ as the probability measures induced by $\mathbf{z}^{(d)} = (\mathbf{x}^{(d)}, \mathbf{y}^{(d)})$ and $\hat{\mathbf{z}}^{(d)} = (\mathbf{x}^{(d)}, \hat{\mathbf{y}}^{(d)})$, respectively. Similarly, let $\mu^{(n)}$ and $\hat{\mu}^{(n)}$ denote the corresponding empirical measures induced by $\mathbf{Z}^{(d)} = \{\mathbf{z}^{(t,d)}\}_{t=1}^{n}$ and $\hat{\mathbf{Z}}^{(d)} = \{\hat{\mathbf{z}}^{(t,d)}\}_{t=1}^{n}$, respectively. For $v \in \mathbb{S}^{T-1}$ and any Borel set $A \subseteq \mathbb{R}$, define

$$\mu_v^{(n)}(A) \coloneqq n^{-1} \sum_{t=1}^{n} \delta_{v^\top \mathbf{z}^{(t,d)}}(A) = n^{-1} \sum_{t=1}^{n} \mathbb{I}_A(v^\top \mathbf{z}^{(t,d)}) \tag{1}$$

$$\hat{\mu}_v^{(n)}(A) \coloneqq n^{-1} \sum_{t=1}^{n} \delta_{v^\top \hat{\mathbf{z}}^{(t,d)}}(A) = n^{-1} \sum_{t=1}^{n} \mathbb{I}_A(v^\top \hat{\mathbf{z}}^{(t,d)}). \tag{2}$$

For any measurable continuity set A such that $\mu_v(\partial A) = 0$, by the strong law of large numbers,

$$\mu_v^{(n)}(A) = n^{-1} \sum_{t=1}^{n} \mathbb{I}_A(v^\top \mathbf{z}^{(t,d)}) \longrightarrow \mathbb{E}[\mathbb{I}_A(v^\top \mathbf{z}^{(d)})] = \mu(\{\mathbf{z} : v^\top \mathbf{z} \in A\}) = \mu_v(A). \tag{3}$$

Similarly, for every continuity set A of $\hat{\mu}_v$, $\hat{\mu}_v^{(n)}(A) \longrightarrow \hat{\mu}_v(A)$. Hence, $\mu_v^{(n)} \Rightarrow \mu_v$, and $\hat{\mu}_v^{(n)} \Rightarrow \hat{\mu}_v$.

Therefore, the empirical Cramér–Wold discrepancy converges to the population version of Cramér–Wold discrepancy. Hence, if $D^2_{\mathrm{CW}}(\mathbf{Z}^{(d)}, \hat{\mathbf{Z}}^{(d)}) = 0$ in the large-sample limit, then $D^2_{\mathrm{CW}}(\mu, \hat{\mu}) = 0$. By Lemma 1, $\mu = \hat{\mu}$. Thus, $P_{\mathbf{z}^{(d)}} = P_{\hat{\mathbf{z}}^{(d)}}$.

Since the fractional differencing transform $(1 - B)^d$ is deterministic and invertible on finite-length trajectories, $P_{\mathbf{X},\mathbf{Y}} = P_{\mathbf{X},\hat{\mathbf{Y}}}$.

By the disintegration theorem, there exist regular conditional probability measures $P_{\mathbf{Y}|\mathbf{X}=\mathbf{x}}$ and $P_{\hat{\mathbf{Y}}|\mathbf{X}=\mathbf{x}}$ such that, for all measurable sets $A \subseteq \mathbb{R}^H$ and $B \subseteq \mathbb{R}^F$,

$$P_{\mathbf{X},\mathbf{Y}}(A \times B) = \int_A P_{\mathbf{Y}|\mathbf{X}=\mathbf{x}}(B)\, dP_{\mathbf{X}}(\mathbf{x}), \quad P_{\mathbf{X},\hat{\mathbf{Y}}}(A \times B) = \int_A P_{\hat{\mathbf{Y}}|\mathbf{X}=\mathbf{x}}(B)\, dP_{\mathbf{X}}(\mathbf{x}). \tag{4}$$

Since $P_{\mathbf{X},\mathbf{Y}} = P_{\mathbf{X},\hat{\mathbf{Y}}}$, the uniqueness of regular conditional probabilities gives

$$P_{\mathbf{Y}|\mathbf{X}=\mathbf{x}} = P_{\hat{\mathbf{Y}}|\mathbf{X}=\mathbf{x}}, \quad P_{\mathbf{X}}\text{-a.e. } \mathbf{x}. \tag{5}$$

This completes the proof. □

A.3 Closed-form expression of D^2_{CW}

For a smoothing parameter $\gamma > 0$, chosen according to Silverman's rule of thumb as $\gamma_n = (4/3n)^{2/5}$ [13], the empirical Cramér–Wold discrepancy admits the following closed-form expression [3]:

$$\begin{aligned} D^2_{\mathrm{CW}}(\mathbf{Z}^{(d)}, \hat{\mathbf{Z}}^{(d)}) = (2n^2\sqrt{\pi\gamma})^{-1} \Bigg[& \sum_{t,t'=H}^{N} \varphi_T\left(\frac{\|\mathbf{z}^{(t,d)} - \mathbf{z}^{(t',d)}\|_2^2}{4\gamma}\right) \\ & + \sum_{t,t'=1}^{n} \varphi_T\left(\frac{\|\hat{\mathbf{z}}^{(t,d)} - \hat{\mathbf{z}}^{(t',d)}\|_2^2}{4\gamma}\right) - 2\sum_{t,t'=1}^{n} \varphi_T\left(\frac{\|\mathbf{z}^{(t,d)} - \hat{\mathbf{z}}^{(t',d)}\|_2^2}{4\gamma}\right)\Bigg]. \end{aligned} \tag{6}$$

For sufficiently large T, $\varphi_T(s) \approx (1 + 4s/(2T-3))^{-1/2}$, which allows D^2_{CW} to be computed efficiently using only pairwise squared Euclidean distances.

B Reproducibility

B.1 Datasets

We evaluate our method ProDA on a diverse collection of benchmark forecasting datasets spanning energy, weather, finance, and traffic domains.

- ETT (Electricity Transformer Temperature) [20]: The ETT dataset contains seven transformer-related variables collected from two regions between July 2016 and July 2018. Variants with suffix "h" correspond to hourly observations, while "m" denotes 15-minute intervals. The suffixes "1" and "2" indicate different geographical regions.
- Weather [18]: The Weather dataset contains 21 meteorological variables recorded every 10 minutes during 2020 at the Max Planck Biogeochemistry Institute weather station.
- ECL (Electricity Consuming Load) [6]: The ECL dataset consists of hourly electricity consumption measurements from 321 clients between 2012 and 2014.
- Exchange [5]: The Exchange dataset contains daily exchange rates from 1990 to 2016 for eight countries, including Australia, the United Kingdom, Canada, Switzerland, China, Japan, New Zealand, and Singapore.
- PEMS [7]: The PEMS dataset contains traffic flow measurements from the California highway system aggregated at 5-minute intervals. We use the PEMS08 subset in our experiments.

We follow the standard preprocessing and train-validation-test split protocols used in prior forecasting studies [20, 18, 8]. Specifically, the ETT datasets are split into training, validation, and test sets using a 12/4/4-month partition, while all other datasets follow a 7:1:2 ratio. Additional dataset statistics are summarized in Table 1.

B.2 Baseline models

We evaluate ProDA using four representative state-of-the-art forecasting models spanning Transformer- and MLP-based architectures. For all experiments, we utilize the official implementations.

- iTransformer [8][1]: iTransformer treats each variable as a token and applies Transformer attention across the channel dimension to explicitly model inter-variable dependencies.

[1] https://github.com/thuml/iTransformer

Table 1: Datasets Statistics Summary.

Dataset	p	Forecast length	Train / validation / test	Frequency	Domain
ETTh1	7	96, 192, 336, 720	8545 / 2881 / 2881	Hourly	Electricity
ETTh2	7	96, 192, 336, 720	8545 / 2881 / 2881	Hourly	Electricity
ETTm1	7	96, 192, 336, 720	34465 / 11521 / 11521	15 min	Electricity
ETTm2	7	96, 192, 336, 720	34465 / 11521 / 11521	15 min	Electricity
Weather	21	96, 192, 336, 720	36792 / 5271 / 10540	10 min	Weather
ECL	321	96, 192, 336, 720	18317 / 2633 / 5261	Hourly	Electricity
Exchange	8	96, 192, 336, 720	5120 / 665 / 1422	1 day	Exchange Rates
PEMS08	170	12, 24, 48, 96	10690 / 3548 / 265	5min	Transportation

- PatchTST [9][2]: PatchTST divides time series into temporal patches and applies a Transformer encoder to capture long-range temporal dependencies efficiently.
- TimeMixer [17][3]: TimeMixer captures multi-scale temporal patterns through hierarchical decomposition and scale-wise feature mixing.
- DLinear [19][4]: DLinear decomposes time series into trend and seasonal components and models each component using lightweight linear layers.

B.3 Comparison Objective Functions

We compare ProDA with recent objective-centric baselines, each of which imposes a different structural view on forecast trajectories, complementing the discussion in Table **??**. We use the official implementations of all objectives and follow the hyperparameter settings recommended by the original authors.

- AliO [10][5]: AliO compares forecasts and observations in both the time and frequency domains to encourage consistency across complementary representations.
- PSLoss [4][6]: PSLoss partitions the forecast horizon into local patches and compares patch-level statistics to emphasize local structure beyond coordinate-wise errors.
- Time-o1 [16][7]: Time-o1 is a recent objective-centric forecasting framework included as an additional comparator.
- DBLoss [12][8]: DBLoss decomposes trajectories into trend and seasonal components and penalizes decomposition-aware forecasting errors.
- FreDF [15][9]: FreDF measures discrepancies in the frequency domain to preserve the spectral structure of the target sequence.
- DistDF [14][10]: DistDF aligns observed and predicted futures using a Wasserstein-based discrepancy over joint trajectory representations.

B.4 Implementation Details

- Our implementation code is provided in the supplementary materials.
- All experiments are conducted on an NVIDIA RTX A6000 GPU using implementations in `PyTorch`.

[2] https://github.com/yuqinie98/PatchTST
[3] https://github.com/kwuking/TimeMixer
[4] https://github.com/cure-lab/LTSF-Linear
[5] https://github.com/eai-lab/AliO
[6] https://github.com/Dilfiraa/PS_Loss
[7] https://github.com/Master-PLC/Time-o1
[8] https://github.com/decisionintelligence/DBLoss
[9] https://github.com/Master-PLC/FreDF
[10] https://github.com/Master-PLC/DistDF

- Training is conducted using the Adam optimizer [2] with an initial learning rate of 0.01. We also employ the `ReduceLROnPlateau` scheduler from `PyTorch`, with a patience of 5 and a reduction factor of 0.1.
- All methods are trained for 100 epochs with a batch size of 512, halved adaptively (to a minimum of 64) if out-of-memory (OOM) issues occur.
- For all methods, the stopping criterion is the validation loss.
- We follow the comprehensive time series forecasting benchmark TFB [11] to set the historical input length H.

Table 2: Hyperparameter search space for each neural network architecture. Bold values denote the configurations used in our experiments.

Hyperparameter	iTransformer [8]	PatchTST [9]	TimeMixer [17]	DLinear [19]
`d_model`	{**128**, 256, 512}	{**32**, 64, 128}	{**16**, 32, 64}	-
`d_ff`	{**128**, 256, 512}	{**128**, 256, 512}	{**32**, 64, 128}	-
`e_layers`	{**2**, 3, 4}	{2, **3**, 4}	{**2**, 3, 4}	-
`moving_avg`	-	-	**25**	**25**

Table 3: Hyperparameter search space for ProDA.

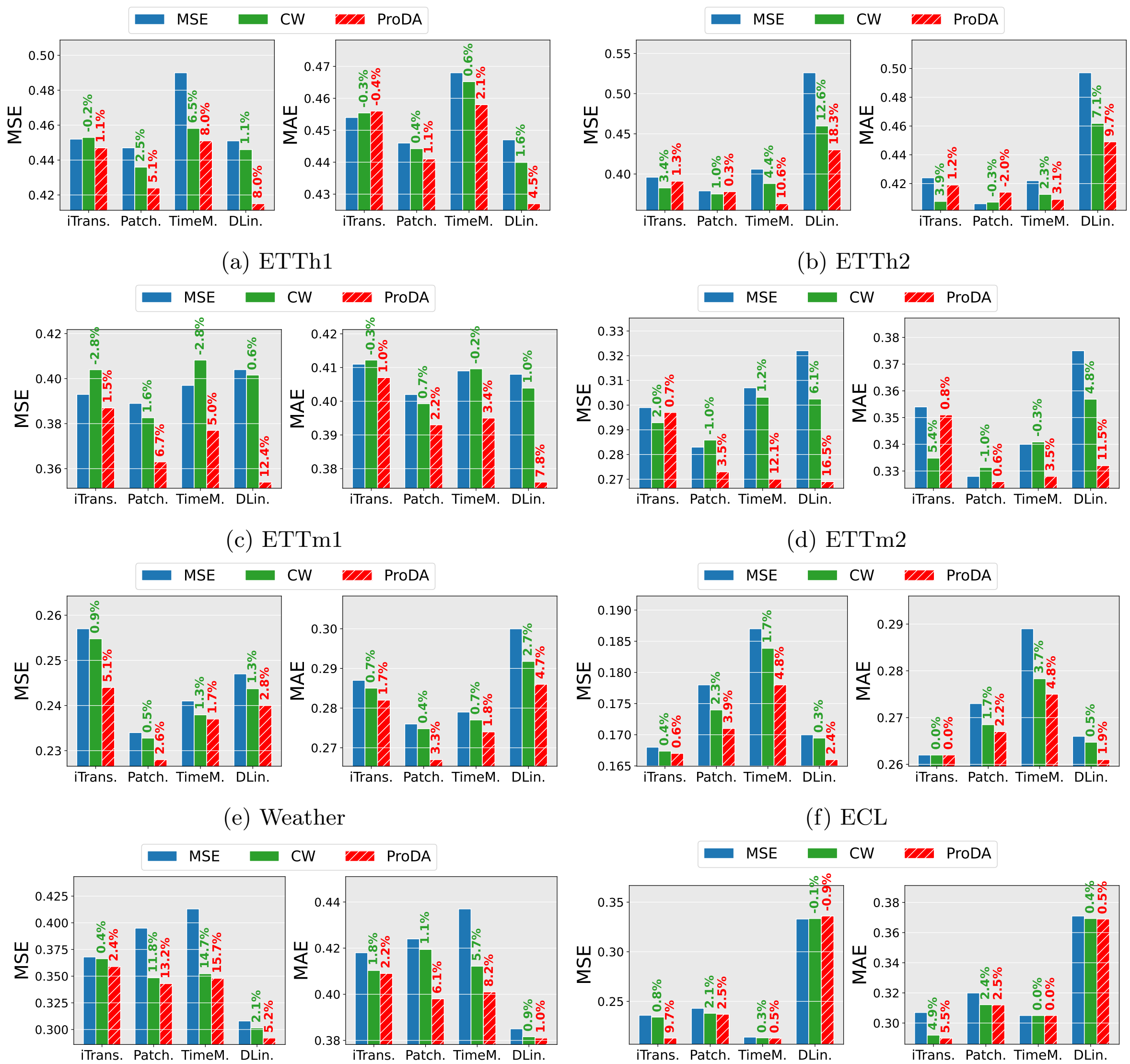
Hyperparameter	ProDA
λ	{0, 0.01, **0.02**, 0.05, 0.1, 0.2, 0.5, 1}
d	{0.05, 0.1, **0.2**, 0.3, 0.4, 0.5}

Hyperparameters. We construct compact hyperparameter search spaces based on the commonly used configurations in the original implementations of each baseline model. For all datasets, we use the same selected configuration for each baseline model without dataset-specific tuning, enabling a consistent evaluation of ProDA across different forecasting settings. Bold values in Tables 2 and 3 denote the final configurations used in our experiments.

C Additional Experimental Results

C.1 RQ2: Ablation Study

In this subsection, we provide the detailed ablation study results for the eight datasets we used.



(a) ETTh1 (b) ETTh2

(c) ETTm1 (d) ETTm2

(e) Weather (f) ECL

(g) Exchange (h) PEMS08

Figure 1: Ablation study of the components of ProDA on the ETTh1 and ETTm1 datasets.

RQ2. Across these eight datasets, the full ProDA objective improves average MSE over the plain MSE objective in 31 of 32 dataset–baseline model cases and improves or matches average MAE in 30 of 32 cases. Compared with using the Cramér–Wold term alone, the full objective yields larger MSE gains in 28 of 32 cases and larger or equal MAE gains in 28 of 32 cases, indicating that fractional differencing provides a meaningful additional benefit beyond distribution alignment alone. The largest MSE gain appears on ETTh2 with DLinear (18.3%), while the largest MAE gain appears on ETTm2 with DLinear (11.5%). Overall, the benefit is strongest for the simpler baseline models, especially TimeMixer and DLinear, and on the more non-stationary datasets such as ETTh2, ETTm2, and Exchange.

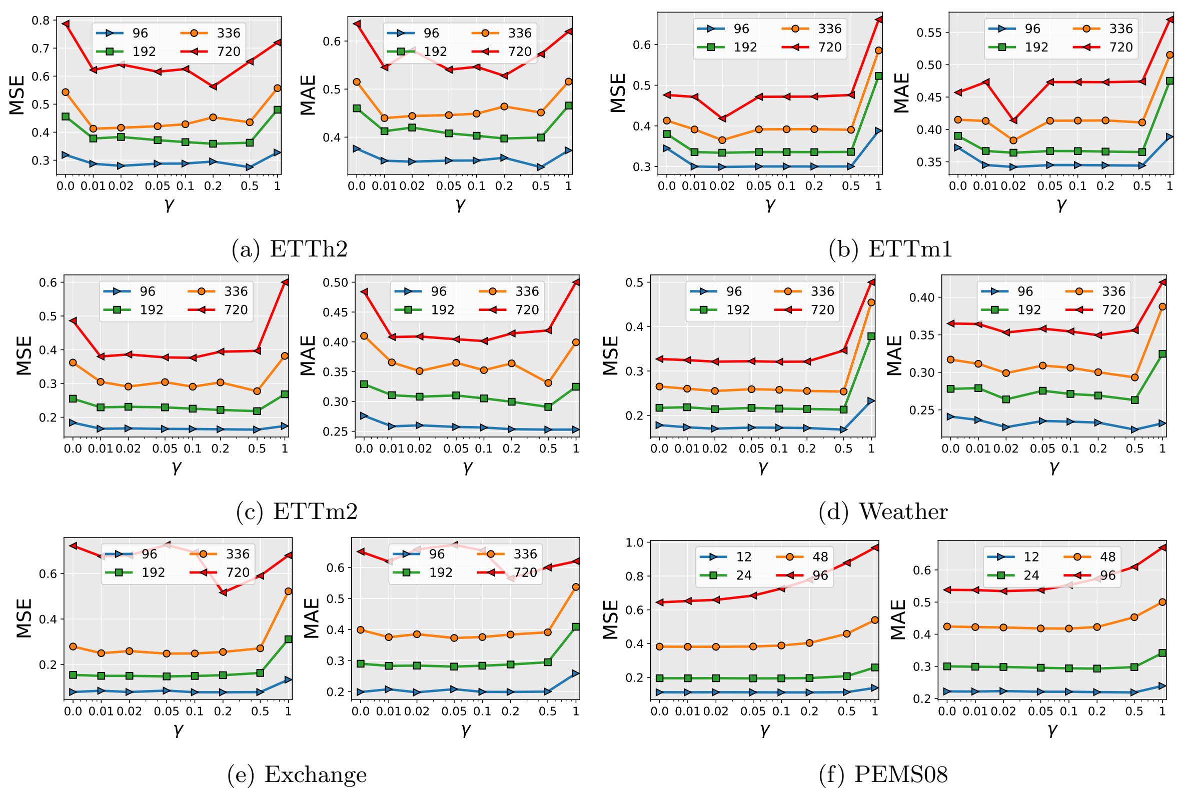
C.2 RQ3: Zero-shot forecasting performance

In this subsection, we provide the detailed zero-shot forecasting performance in the ETT datasets.

RQ3. The detailed horizon-wise results reinforce the average trends reported in the main paper: across all 192 source–target–horizon–baseline model comparisons, ProDA improves MSE in 163 cases and MAE in 148 cases. The gains are especially consistent for DLinear and TimeMixer; DLinear improves MSE in 47 of 48 detailed comparisons and improves or matches MAE in all 48, while TimeMixer improves MSE in 45 of 48 and MAE in 42 of 48. The improvement rate is highest at horizon 192 for MSE, where ProDA improves 46 of 48 comparisons, and remains strong even at horizon 720, where it still improves 37 of 48 MSE comparisons and 36 of 48 MAE comparisons. These detailed results show that the transfer benefit of ProDA is not confined to a small subset of forecast lengths or source–target pairs.

C.3 RQ4: Hyperparameter sensitivity

In this subsection, we provide the hyperparameter sensitivity across six datasets, including ETTh2, ETTm1, ETTm2, Weather, Exchange, and PEMS08.



(a) ETTh2 (b) ETTm1

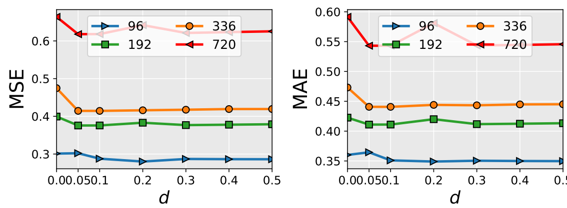
(c) ETTm2 (d) Weather

(e) Exchange (f) PEMS08

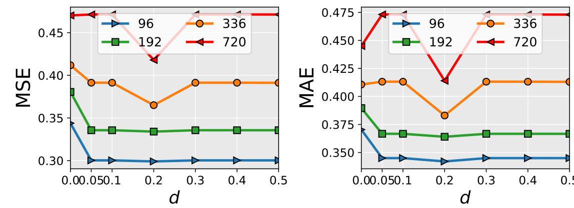
Figure 2: Sensitivity analysis of the hyperparameter γ of ProDA in DLinear model.

Table 4: Zero-shot forecasting results on ETT datasets.

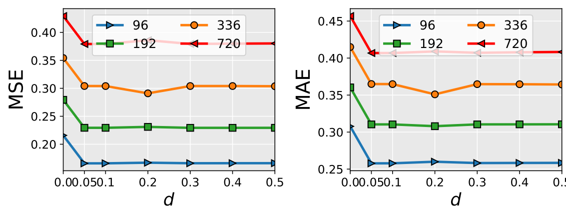
model		iTransformer [8] (2024)				PatchTST [9] (2023)				TimeMixer [17] (2024)				DLinear [19] (2023)			
loss		Original		+ **ProDA**		Original		+ **ProDA**		Original		+ **ProDA**		Original		+ **ProDA**	
metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Forecasting horizon = 96																	
ETTh1	ETTh2	0.297	0.344	0.297	0.344	0.301	0.345	0.301	0.345	0.303	0.353	0.287	0.345	0.321	0.378	0.306	0.370
	ETTm1	0.931	0.601	0.931	0.601	0.885	0.591	0.886	0.592	0.773	0.558	0.739	0.548	0.732	0.552	0.718	0.548
	ETTm2	0.229	0.313	0.229	0.313	0.217	0.303	0.217	0.303	0.216	0.302	0.206	0.298	0.247	0.351	0.240	0.345
ETTh2	ETTh1	0.590	0.528	0.584	0.521	0.501	0.466	0.570	0.518	0.821	0.589	0.619	0.531	0.422	0.427	0.417	0.427
	ETTm1	1.139	0.669	1.055	0.640	0.928	0.591	0.736	0.534	0.987	0.604	0.788	0.556	0.775	0.561	0.760	0.556
	ETTm2	0.241	0.326	0.241	0.323	0.225	0.309	0.203	0.295	0.225	0.308	0.204	0.297	0.249	0.351	0.224	0.327
ETTm1	ETTh1	0.778	0.580	0.668	0.553	0.617	0.515	0.542	0.486	0.903	0.667	0.599	0.538	0.582	0.500	0.471	0.449
	ETTh2	0.364	0.398	0.333	0.384	0.358	0.386	0.349	0.386	0.372	0.403	0.325	0.385	0.356	0.405	0.311	0.369
	ETTm2	0.205	0.281	0.184	0.270	0.198	0.270	0.182	0.265	0.201	0.278	0.180	0.265	0.215	0.306	0.185	0.277
ETTm2	ETTh1	0.670	0.547	0.836	0.615	0.556	0.506	0.587	0.520	0.859	0.617	0.744	0.603	0.564	0.492	0.434	0.441
	ETTh2	0.360	0.388	0.364	0.409	0.332	0.367	0.295	0.357	0.354	0.387	0.351	0.392	0.312	0.366	0.294	0.356
	ETTm1	0.599	0.486	0.467	0.447	0.535	0.451	0.424	0.420	0.519	0.463	0.425	0.420	0.489	0.442	0.360	0.389
Forecasting horizon = 192																	
ETTh1	ETTh2	0.376	0.392	0.358	0.395	0.381	0.394	0.347	0.389	0.384	0.397	0.346	0.384	0.423	0.441	0.379	0.414
	ETTm1	0.874	0.592	0.930	0.619	0.855	0.596	0.756	0.578	0.791	0.557	0.745	0.560	0.759	0.573	0.716	0.553
	ETTm2	0.289	0.346	0.281	0.352	0.277	0.337	0.275	0.345	0.277	0.336	0.262	0.332	0.333	0.408	0.299	0.380
ETTh2	ETTh1	0.648	0.555	0.623	0.556	0.609	0.537	0.528	0.491	1.404	0.764	0.741	0.613	0.473	0.463	0.437	0.441
	ETTm1	1.012	0.636	1.005	0.645	0.965	0.613	0.742	0.544	2.836	0.953	1.044	0.640	0.804	0.590	0.717	0.553
	ETTm2	0.299	0.355	0.287	0.356	0.293	0.349	0.260	0.333	0.362	0.381	0.273	0.342	0.371	0.433	0.309	0.388
ETTm1	ETTh1	0.738	0.570	0.606	0.525	0.695	0.571	0.528	0.478	0.877	0.632	0.630	0.538	0.578	0.504	0.481	0.460
	ETTh2	0.454	0.442	0.395	0.417	0.457	0.440	0.402	0.419	0.545	0.487	0.375	0.411	0.449	0.456	0.388	0.419
	ETTm2	0.261	0.315	0.240	0.308	0.258	0.312	0.235	0.300	0.282	0.334	0.229	0.299	0.286	0.357	0.254	0.331
ETTm2	ETTh1	0.855	0.619	1.011	0.684	0.641	0.553	0.542	0.495	0.785	0.593	0.739	0.577	0.592	0.511	0.483	0.469
	ETTh2	0.435	0.437	0.422	0.441	0.414	0.418	0.389	0.415	0.416	0.423	0.381	0.415	0.410	0.425	0.360	0.395
	ETTm1	0.650	0.517	0.560	0.505	0.579	0.488	0.499	0.467	0.551	0.474	0.461	0.446	0.491	0.449	0.397	0.407
Forecasting horizon = 336																	
ETTh1	ETTh2	0.422	0.425	0.383	0.418	0.424	0.431	0.384	0.421	0.426	0.433	0.382	0.415	0.493	0.486	0.451	0.461
	ETTm1	0.922	0.614	0.935	0.630	0.855	0.608	0.796	0.607	0.799	0.587	0.761	0.581	0.756	0.580	0.733	0.571
	ETTm2	0.351	0.382	0.336	0.385	0.337	0.372	0.331	0.377	0.332	0.368	0.319	0.366	0.409	0.454	0.386	0.434
ETTh2	ETTh1	0.674	0.567	0.761	0.624	0.722	0.595	0.875	0.665	0.763	0.593	0.715	0.576	0.502	0.480	0.454	0.457
	ETTm1	0.967	0.630	1.076	0.673	0.950	0.621	0.748	0.554	1.542	0.766	0.757	0.572	0.781	0.592	0.704	0.559
	ETTm2	0.355	0.385	0.348	0.388	0.359	0.388	0.319	0.368	0.380	0.396	0.321	0.368	0.464	0.488	0.396	0.442
ETTm1	ETTh1	0.960	0.664	0.644	0.553	0.669	0.565	0.498	0.468	0.957	0.711	0.646	0.576	0.598	0.522	0.446	0.454
	ETTh2	0.573	0.509	0.455	0.456	0.485	0.467	0.383	0.415	0.562	0.519	0.398	0.431	0.510	0.498	0.404	0.434
	ETTm2	0.342	0.367	0.302	0.345	0.314	0.347	0.287	0.335	0.330	0.359	0.280	0.331	0.368	0.414	0.297	0.359
ETTm2	ETTh1	1.131	0.721	1.080	0.706	0.664	0.568	0.588	0.547	0.907	0.651	1.078	0.742	0.612	0.531	0.512	0.487
	ETTh2	0.530	0.496	0.468	0.468	0.446	0.450	0.404	0.428	0.441	0.449	0.439	0.457	0.497	0.488	0.405	0.432
	ETTm1	0.768	0.572	0.603	0.524	0.621	0.531	0.468	0.437	0.708	0.547	0.531	0.480	0.512	0.470	0.416	0.420
Forecasting horizon = 720																	
ETTh1	ETTh2	0.426	0.441	0.426	0.442	0.427	0.442	0.426	0.452	0.434	0.448	0.421	0.445	0.618	0.563	0.611	0.557
	ETTm1	0.830	0.596	0.832	0.597	0.897	0.618	0.802	0.616	0.822	0.611	0.930	0.682	0.775	0.607	0.758	0.603
	ETTm2	0.438	0.425	0.439	0.426	0.444	0.431	0.415	0.424	0.430	0.420	0.449	0.446	0.604	0.553	0.605	0.548
ETTh2	ETTh1	0.722	0.606	0.712	0.601	0.674	0.581	0.687	0.603	1.498	0.809	0.826	0.662	0.541	0.532	0.490	0.508
	ETTm1	0.913	0.627	0.910	0.627	0.899	0.624	0.752	0.570	3.472	1.117	1.005	0.635	0.808	0.626	0.742	0.599
	ETTm2	0.449	0.433	0.449	0.433	0.446	0.431	0.399	0.415	0.577	0.512	0.419	0.424	0.773	0.630	0.708	0.593
ETTm1	ETTh1	1.152	0.752	0.668	0.575	0.731	0.615	0.524	0.500	1.413	0.864	0.703	0.582	0.590	0.547	0.482	0.499
	ETTh2	0.642	0.552	0.478	0.482	0.496	0.486	0.448	0.462	0.574	0.527	0.423	0.450	0.585	0.550	0.533	0.529
	ETTm2	0.455	0.429	0.387	0.397	0.417	0.406	0.378	0.391	0.431	0.411	0.370	0.388	0.521	0.502	0.492	0.486
ETTm2	ETTh1	0.989	0.687	1.071	0.717	0.800	0.644	0.944	0.709	4.074	1.151	1.053	0.750	0.597	0.551	0.478	0.493
	ETTh2	0.496	0.488	0.499	0.493	0.452	0.463	0.426	0.455	0.506	0.496	0.401	0.452	0.541	0.527	0.416	0.454
	ETTm1	0.809	0.598	0.679	0.570	0.769	0.599	0.559	0.491	1.302	0.730	0.551	0.498	0.548	0.495	0.446	0.438



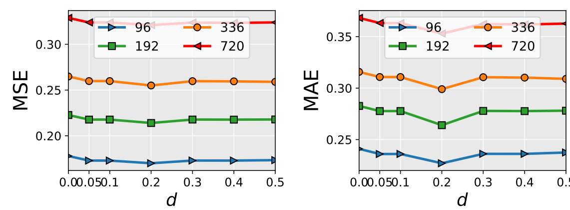
(a) ETTh2



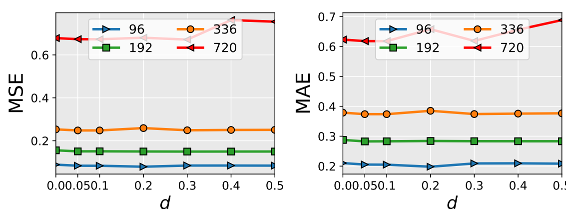
(b) ETTm1



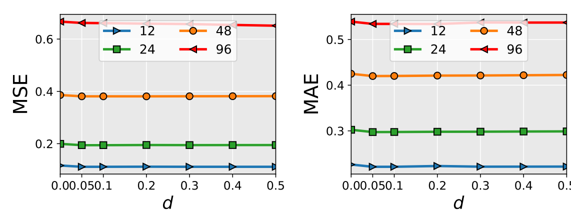
(c) ETTm2



(d) Weather



(e) Exchange



(f) PEMS08

Figure 3: Sensitivity analysis of the hyperparameter differencing order d of ProDA in DLinear model.

RQ4. The supplementary sensitivity plots further support the robustness of ProDA. For the smoothing parameter γ, most datasets attain their best or near-best performance in a moderate range roughly between 0.01 and 0.2, whereas the extreme choice $\gamma = 1$ often causes a visible degradation, especially at the longest horizons on ETTh2, ETTm1, ETTm2, Weather, and PEMS08. For the differencing order d, the curves are generally flatter: values in the range $[0.05, 0.3]$ are consistently competitive across all datasets, with ETTm1 showing its clearest optimum around $d = 0.2$ and Exchange remaining almost unchanged across a wider range. Overall, these supplementary panels indicate that the method does not depend on highly fragile hyperparameter tuning, although moderate values are consistently safer than the extremes.

C.4 RQ5: Computational Efficiency

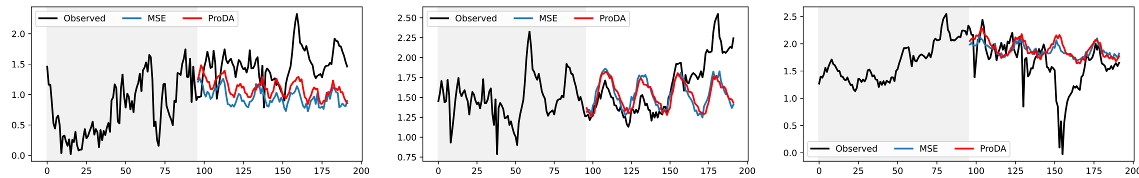
Table 5: Training time (s/epoch) comparison. Δ denotes the training time increment of PRODA over MSE.

dataset	ETTh1			ETTh2			ETTm1			ETTm2			Weather			Exchange			ECL			PEMS08		
metric	MSE	ProDA	Δ	MSE	ProDA	Δ	MSE	ProDA	Δ	MSE	ProDA	Δ	MSE	ProDA	Δ	MSE	ProDA	Δ	MSE	ProDA	Δ	MSE	ProDA	Δ
iTransformer																								
96 (12)	1.48	1.64	0.16	1.49	1.62	0.13	3.82	3.94	0.13	3.83	4.12	0.29	18.45	21.86	3.41	1.36	1.45	0.08	17.72	26.01	8.30	14.72	16.31	1.59
192 (24)	1.52	1.82	0.30	1.52	1.80	0.28	4.06	4.25	0.19	4.09	4.14	0.04	18.85	23.11	4.26	1.36	1.47	0.11	21.11	33.85	12.74	14.97	15.91	0.94
336 (48)	1.57	1.81	0.24	1.58	1.76	0.18	4.41	4.49	0.08	4.37	4.59	0.21	19.26	22.56	3.30	1.39	1.60	0.21	26.34	44.64	18.29	15.33	16.15	0.82
720 (96)	1.67	1.74	0.07	1.67	1.76	0.09	5.40	5.78	0.38	5.29	5.35	0.06	20.98	25.36	4.38	1.40	1.54	0.14	39.98	52.38	12.40	16.16	18.31	2.16
Avg.	1.56	1.75	0.19	1.56	1.73	0.17	4.42	4.62	0.20	4.39	4.55	0.15	19.39	23.22	3.84	1.38	1.51	0.14	26.29	39.22	12.93	15.30	16.67	1.38
DLinear																								
96 (12)	1.34	1.58	0.23	1.34	1.64	0.30	3.04	5.56	2.51	3.02	3.46	0.43	17.73	21.95	4.22	1.25	1.55	0.30	9.67	18.89	9.22	12.71	15.34	2.63
192 (24)	1.37	1.60	0.23	1.37	1.63	0.25	2.94	4.02	1.07	3.29	3.69	0.40	18.15	21.30	3.16	1.27	1.36	0.09	12.87	25.01	12.14	12.91	15.85	2.94
336 (48)	1.42	1.62	0.20	1.43	1.63	0.21	3.73	3.92	0.19	3.60	4.01	0.41	18.64	22.50	3.86	1.29	1.39	0.10	18.38	36.54	18.16	13.37	15.89	2.52
720 (96)	1.53	1.70	0.17	1.52	1.71	0.19	4.67	4.84	0.16	4.68	4.70	0.02	20.32	23.31	2.98	1.33	1.44	0.12	30.36	43.40	13.04	14.10	16.20	2.09
Avg.	1.42	1.63	0.21	1.42	1.65	0.24	3.60	4.58	0.98	3.65	3.96	0.32	18.71	22.27	3.56	1.29	1.44	0.15	17.82	30.96	13.14	13.27	15.82	2.54

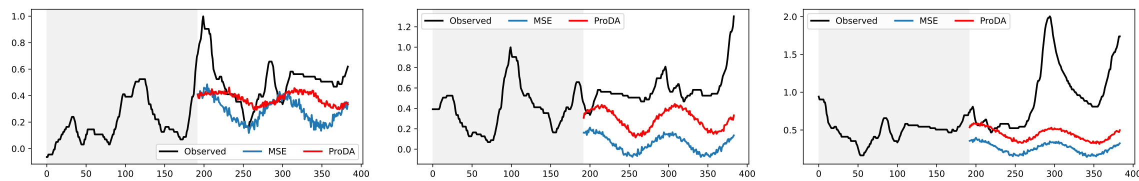
RQ5. Table 5 reports the training time per epoch across forecasting horizons. Overall, ProDA introduces modest overhead on most datasets for both iTransformer and DLinear. The average increase is below 0.25 seconds on ETTh1, ETTh2, and Exchange, and remains relatively small on ETTm1 and ETTm2. Larger overheads are observed on Weather and ECL, especially for ECL, where the average increase reaches 12.93 and 13.14 seconds for iTransformer and DLinear, respectively. Nevertheless, the results show that ProDA remains computationally practical while adding manageable training cost.

C.5 Forecasting Visualization

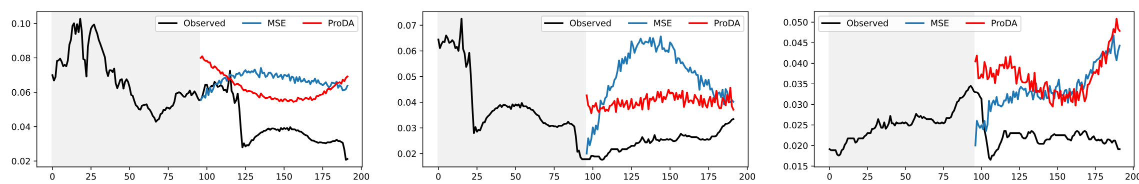
In this section, we visualize forecasting samples from the ETTh1, ETTm2, Weather, ECL, and PEMS08 datasets using iTransformer as the baseline model. Compared with standard point-wise training, predictions generated with ProDA show better alignment with the ground-truth trajectories, capturing temporal directionality, fluctuation dynamics, and overall trends.



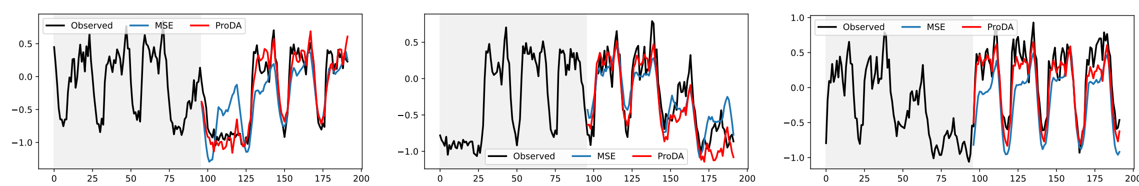
(a) ETTh1



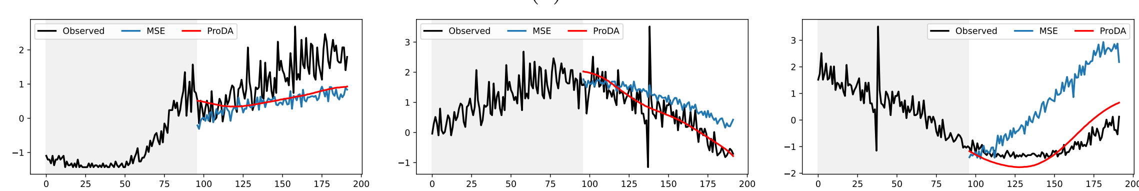
(b) ETTm2



(c) Weather



(d) ECL



(e) PEMS08

Figure 4: Forecasting visualization of ProDA in the iTransformer model.

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